

Windows System Setup Guide for Training Session

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1 Introduction

This document provides step-by-step instructions to set up a Windows system for a training session requiring specific hardware and software configurations. The setup includes Docker Desktop, Kubernetes tools, Python, Git, and Visual Studio Code (VS Code) with Kubernetes extensions.

2 Prerequisites

2.1 Hardware Requirements

- Laptop/Desktop with a high-speed internet connection.
- Minimum 8 GB RAM.
- Minimum 4 CPU cores.
- At least 10 GB of free disk space.

2.2 Software Requirements

- Windows 10 or later.
- Administrative privileges for software installation.
- A GitHub account (<https://github.com>).

3 Step 1: Verify Hardware Requirements

1. Check RAM and CPU:

- Right-click Start menu, select *System*.
- Under *Device specifications*, ensure *Installed RAM* is 8 GB or more.
- Under *Processor*, confirm at least 4 CPU cores (e.g., Intel i5/i7 or AMD Ryzen).

2. Check Disk Space:

- Open *File Explorer*, right-click C: drive, select *Properties*.
- Verify at least 10 GB free space.

3. Internet Connection:

- Test at <https://www.speedtest.net> for a stable, high-speed connection.

4 Step 2: Install Required Software

4.1 1. Install Docker Desktop

1. **Download:** Visit <https://www.docker.com/products/docker-desktop>, download for Windows.
2. **Install:**
 - Run the .exe file, follow the wizard.

- Enable *Hyper-V* or *WSL 2* as prompted.
3. **Enable WSL 2 (if prompted):**
 - Open PowerShell as Administrator, run: `wsl -install`.
 - Restart system if required.
 4. **Verify:** Run `docker -version` and `docker run hello-world` in PowerShell.

4.2 2. Install kubectl CLI

1. **Download:** Get the latest `kubectl.exe` from <https://kubernetes.io/docs/tasks/tools/install-kubectl-windows>.
2. **Install:**
 - Move `kubectl.exe` to `C:\Program Files\kubectl`.
 - Add to PATH via *System > Advanced system settings > Environment Variables > Path*.
3. **Verify:** Run `kubectl version -client` in PowerShell.

4.3 3. Install kind (Kubernetes in Docker)

1. **Download:** Get `kind-windows-amd64` from <https://kind.sigs.k8s.io/docs/user/quick-start>.
2. **Install:**
 - Rename to `kind.exe`, move to `C:\Program Files\kind`.
 - Add to PATH.
3. **Verify:** Run `kind version` in PowerShell.

4.4 4. Install kustomize CLI

1. **Download:** Get `kustomize_..._windows_amd64.tar.gz` from <https://kubectl.docs.kubernetes.io/installation/kustomize>.
2. **Install:**
 - Extract `kustomize.exe`, move to `C:\Program Files\kustomize`.
 - Add to PATH.
3. **Verify:** Run `kustomize version` in PowerShell.

4.5 5. Install Git

1. **Download:** Get Git for Windows from <https://git-scm.com/download/win>.
2. **Install:**
 - Run installer, select *Use Git from the Windows Command Prompt*.
 - Choose preferred editor (e.g., VS Code).
3. **Verify:** Run `git -version`, configure:

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```
4. **Authenticate:** Set up SSH or Personal Access Token (<https://docs.github.com/en/authentication>).

4.6 6. Install Python

1. **Download:** Get Python 3.x from <https://www.python.org/downloads/windows>.
2. **Install:**
 - Run installer, check *Add Python to PATH*.
 - Select *Install Now*.
3. **Verify:** Run `python -version` and `pip -version` in PowerShell.
4. **Optional Virtual Environment:**

```
python -m venv myenv
.\myenv\Scripts\activate
pip install numpy pandas scikit-learn tensorflow
```

4.7 7. Install Visual Studio Code (VS Code)

1. **Download:** Get VS Code from <https://code.visualstudio.com/download>.
2. **Install:** Run installer with default settings.
3. **Install Extensions:**
 - Open VS Code, go to Extensions (Ctrl+Shift+X).
 - Install: *Kubernetes* (Microsoft), *YAML* (Red Hat), *Docker* (Microsoft), *Python* (Microsoft).
4. **Verify:** Confirm extensions are listed in Extensions view.

5 Step 3: Test the Setup

1. **Test Docker and kind:**

```
kind create cluster --name test-cluster
kubectl cluster-info
kind delete cluster --name test-cluster
```

2. **Test Git:** Clone a repository:

```
git clone https://github.com/kubernetes/examples.git
```

3. **Test Python:** Run:

```
python -c "print('Hello, Python!')"
```

4. **Test VS Code:** Open cloned repository:

```
code examples
```

6 Step 4: Additional Notes

- **Podman Alternative:** Install Podman Desktop (<https://podman-desktop.io>) if preferred, but verify `kind` compatibility.

- **Updates:** Keep tools updated using respective commands or installers.
- **Troubleshooting:**
 - Docker issues: Ensure Hyper-V or WSL 2 is enabled.
 - Command failures: Verify tools are in PATH.

7 Step 5: Prepare for the Training Session

- Create a workspace folder (e.g., `C:\Training`).
- Back up important files.
- Ensure stable internet during the session.
- Confirm specific requirements (e.g., Python packages, repositories) with the organizer.